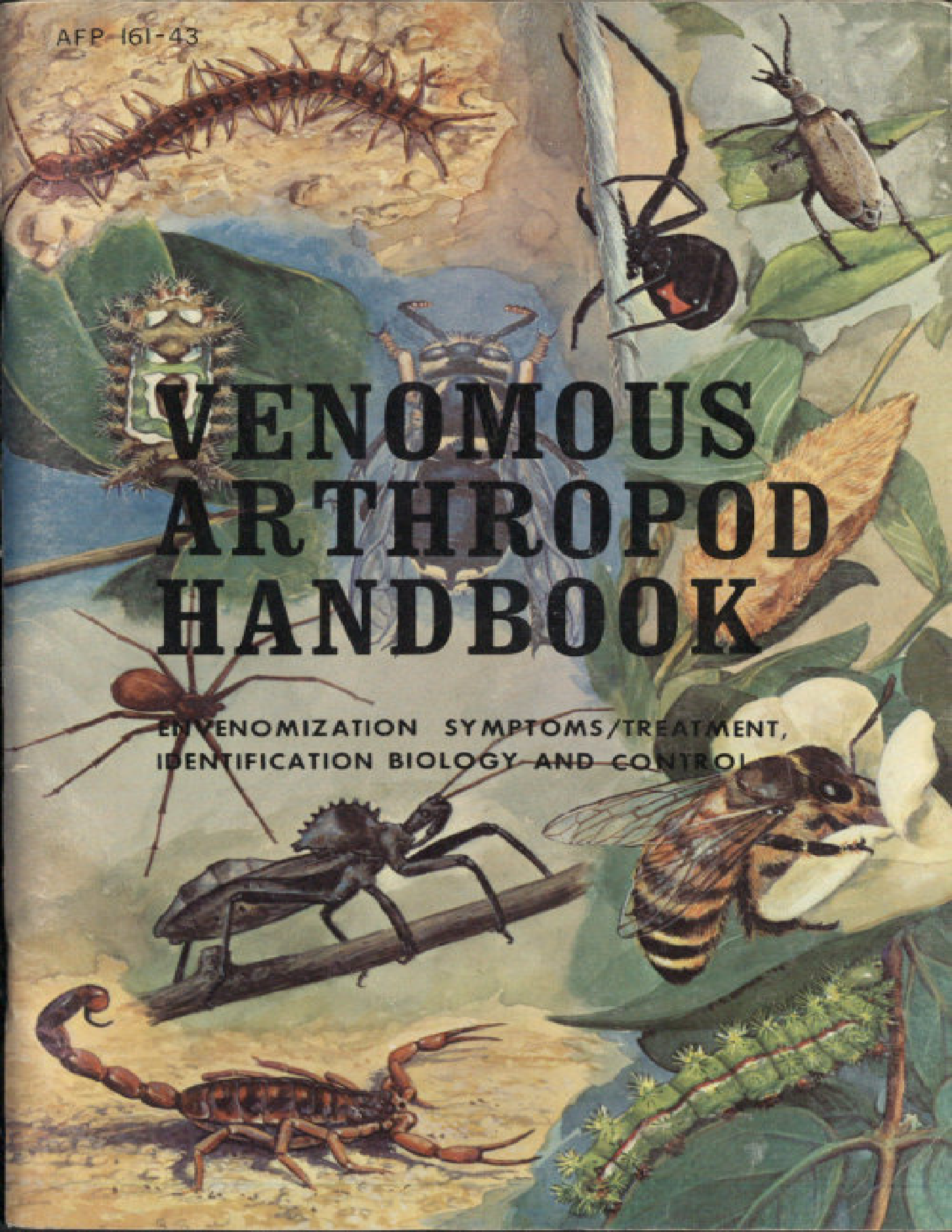




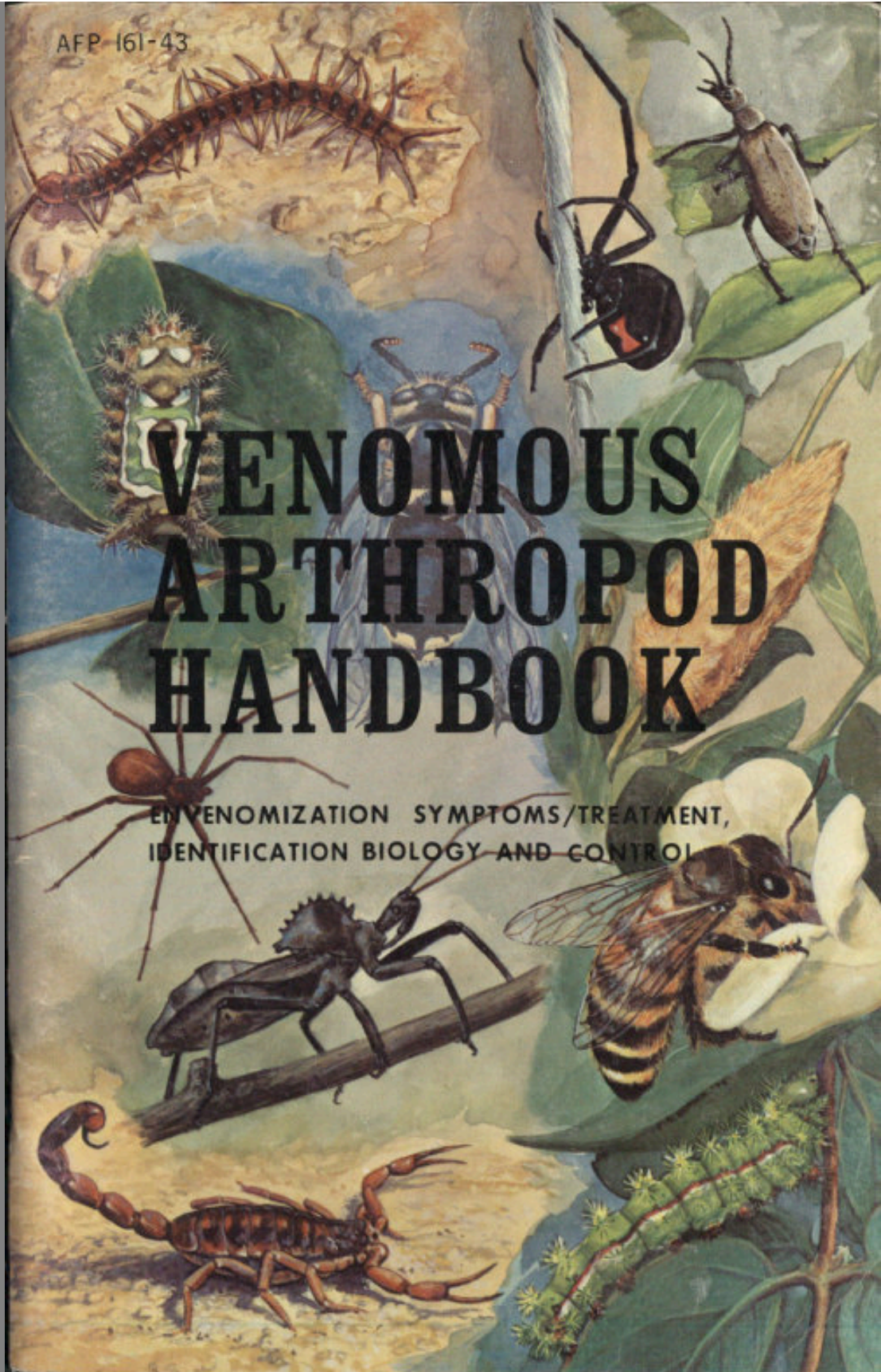
VENOMOUS ARTHROPOD HANDBOOK

ENVENOMIZATION SYMPTOMS/TREATMENT,
IDENTIFICATION BIOLOGY AND CONTROL

AFP 161-43

VENOMOUS ARTHROPOD HANDBOOK

ENVENOMIZATION SYMPTOMS/TREATMENT,
IDENTIFICATION BIOLOGY AND CONTROL



The Project Gutenberg eBook of Venomous arthropod handbook

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Venomous arthropod handbook
envenomization symptoms/treatment, identification, biology
and control

Author: Terry L. Biery

Release date: March 9, 2018 [eBook #56710]

Language: English

Other information and formats: www.gutenberg.org/ebooks/56710

Credits: Produced by Stephen Hutcheson and the Online Distributed Proofreading Team at <http://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK VENOMOUS
ARTHROPOD HANDBOOK ***

VENOMOUS ARTHROPOD HANDBOOK

**Envenomization Symptoms/Treatment, Identification, Biology
and Control**

Terry L. Biery, Captain, USAF, BSC

**Disease Surveillance Branch
Epidemiology Division
USAF School of Aerospace Medicine
Aerospace Medical Division (AFSC)
Brooks Air Force Base, Texas 78235**

For sale by the Superintendent of Documents. U.S. Government Printing Office
Washington, DC. 20402

Stock No. 008-070-00397-0/Catalog No. D 301,35: 161/43
U.S. GOVERNMENT PRINTING OFFICE: 1977 O—251-048

PREFACE

The purpose of this handbook is to provide Air Force medical personnel with a quick reference on the problems caused by the more important venomous arthropods found in the United States. An attempt has been made to include the latest information available in the scientific literature.

The author would like to acknowledge the valuable assistance of Mr. Orville V. Anderson, USAFSAM Photographer; Captain Robert W. Clegern, USAF, BSC, Medical Entomologist; and Lieutenant Colonel John C. Moseley, USAF, MC, Dermatologist.

CONTENTS

<u>INTRODUCTION</u>	1
<u>BITING/PIERCING ARTHROPODS</u>	4
1. <u>Black Widow Spider (<i>Latrodectus mactans</i>)</u>	4
2. <u>Brown Recluse Spider (<i>Loxosceles reclusa</i>)</u>	6
3. <u>Centipedes (Class: Chilopoda)</u>	8
4. <u>Conenose or Kissing Bugs (<i>Triatoma</i> spp.)</u>	10
5. <u>Wheel Bug (<i>Arilus cristatus</i>)</u>	12
<u>STINGING ARTHROPODS</u>	14
1. <u>Ants (Order: Hymenoptera; Family: Formicidae)</u>	14
2. <u>Bees (Order: Hymenoptera; Family: Apidae)</u>	17
3. <u>Scorpions (Class: Arachnida; Order: Scorpionida)</u>	20
4. <u>Velvet Ants (Order: Hymenoptera; Family: Mutillidae)</u>	22
5. <u>Wasps (Order: Hymenoptera; Family: Vespidae)</u>	24
<u>URTICATING/VESICATING ARTHROPODS</u>	26
1. <u>Blister Beetles (Order: Coleoptera; Family: Meloidae)</u>	26
2. <u>Urticating Caterpillars</u>	28
a. <u>Io Moth Caterpillar (<i>Automeris io</i>)</u>	28
b. <u>Puss Caterpillar (<i>Megalopyge opercularis</i>)</u>	30
c. <u>Saddleback Caterpillar (<i>Sibine stimulea</i>)</u>	32
<u>BIBLIOGRAPHY</u>	34
<u>APPENDIX: ARTHROPOD ENVENOMIZATION REACTIONS AND TREATMENT</u>	
	39
<u>Chart 1: Antihistamines</u>	48
<u>Chart 2: Treatment of Anaphylaxis—Shock</u>	49

INTRODUCTION

Although most residents of the United States are aware of the pain and discomfort associated with contacting a venomous arthropod, many are not aware that these arthropods (such as ants, bees, wasps, spiders, and scorpions) are responsible for more human deaths in the United States each year than any other group of venomous animals, including snakes. In fact, a study by Parrish¹⁵ showed that venomous arthropods accounted for more than 65% of the reported human fatalities caused by venomous animals in the United States from 1950 through 1959. His findings are summarized as follows:

<i>Venomous animal group</i>	<i>Human fatalities (1950-59)</i>	
Hymenoptera (bees, wasps, and ants)	229/460 or	49.8%
Poisonous snakes	138/460 or	30.0%
Spiders	65/460 or	14.1%
Scorpions	8/460 or	1.7%
Others	20/460 or	4.4%

It would be incorrect, however, to assume from the above information that a high percentage of arthropod envenomizations are fatal. In actuality, the opposite is true. Of the estimated millions of people envenomized by arthropods in the United States each year, about 25,000 have severe reactions; and of these, approximately 30 result in reported deaths.¹⁹ Therefore, the high number of fatalities caused by venomous arthropods, relative to other venomous animals, can be attributed to the high incidence of arthropod envenomizations rather than the efficient killing power of their venoms. This high-incidence rate is accounted for by the tremendous abundance of venomous arthropods and their ability to live in close proximity to man.

Air Force personnel stationed in the continental United States face a greater risk of contacting a venomous arthropod than the average United States resident because: (1) a large portion of bases are located in the southern half of the United States, where venomous arthropods are most numerous; (2) many facilities (such as radar, communications, security, and missile sites) are located in remote areas where venomous arthropods are not controlled; (3) numerous Air Force training and work activities must be performed outside; and (4) many military structures (such as older wooden buildings, storage buildings, and field training facilities) provide excellent habitats for venomous arthropods. It is important, therefore, that Air Force medical facilities have current information available concerning the most important venomous arthropods in their areas. 2

The venomous arthropods discussed in this handbook were selected for inclusion due to the relative severity of their envenomizations and/or the high frequency of their contact with humans. It should be stressed that other arthropods found in the United States are capable of producing envenomization reactions in humans. Regardless, all venomous arthropods found in the United States can be categorized under one of the three types of envenomization methods described in this handbook: (1) biting/piercing, (2) stinging, and (3) urticating/vesicating. The biting/piercing arthropods inject a toxin through an apparatus associated with or near their mouthparts; and the stinging arthropods, through a stinger located at the posterior end of the abdomen. Urticating and vesicating arthropods are grouped together because they both release their toxins on contact. They differ in that the urticating arthropods usually release their toxin through venomous hairs; and vesicating arthropods carry a toxic substance within their system, which is released through small body openings.

To facilitate the utilization of this handbook, each venomous arthropod presentation has been organized into the following information categories:

- **Identification.** Bringing the arthropod causing an envenomization to the medical facility treating the victim is most helpful. In most cases, an identification must be made before the proper form of treatment can be determined. To assist personnel with identification, the key characteristics 3

and representative color photographs are provided for the venomous arthropods discussed.

- **Distribution.** Most arthropod envenomizations occur during the warm plant-growing season when venomous arthropods are most numerous. Since the Southern United States has the longest growing season, venomous arthropods are abundant for a longer period of time in this region. Also, the moderate climate of the Southern States, with fewer weather extremes, permits a greater variety of venomous arthropods to survive, propagate, and produce multiple generations each year. Specific regional occurrences are indicated.

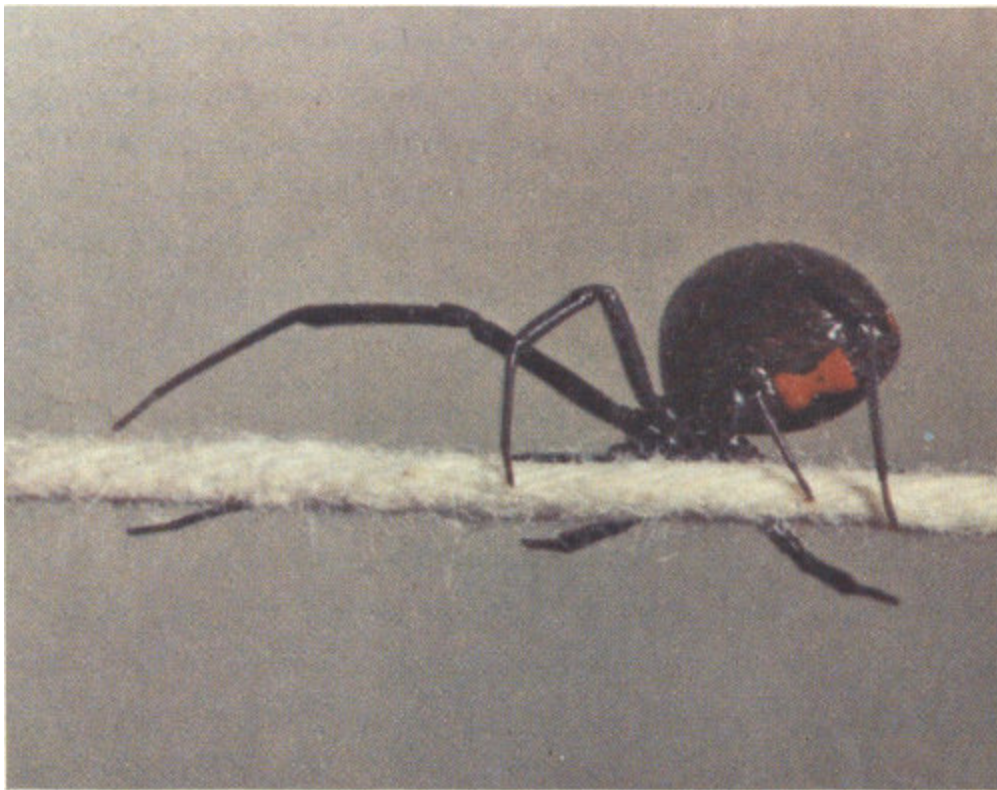
- **Biology/Behavior.** Pertinent biological information is provided to help increase the reader's awareness and understanding of venomous arthropods. These factors are essential in developing a good avoidance and control program.

- **Avoidance/Control.** To assist in preventing envenomization accidents, several means of avoidance are listed for each venomous arthropod. Since the status of insecticide recommendations is constantly changing, few specific recommendations are included. When insecticide control is necessary, an Air Force, County Extension Service, or United States Department of Agriculture entomologist should be consulted for current recommendations. Any insecticide must be applied according to the instructions on the container label.

In addition to the information presented in the text, a table on envenomization reactions and treatment is included as an [appendix](#). The reactions listed in this table are “typical” and do not necessarily represent the entire range of reactions which might occur.

BITING/PIERCING ARTHROPODS

1. BLACK WIDOW SPIDER (*Latrodectus mactans*)



Identification. The female black widow has a shiny black body, approximately 1.5 cm (0.6 in) long, and usually a red hourglass-shaped marking on the underside of her globose abdomen. On some individuals the distinct hourglass marking is replaced with several triangles or spots or an irregular longitudinal blotch. She has slim black legs with a span of 4 cm (1.5 in). The male is considerably smaller than the female, usually a patterned brown color, and is not a threat to man.

Distribution. *Latrodectus mactans* is found in every state except Alaska; however, most reported human fatalities have occurred in the Southeastern [14](#), [15](#) States.

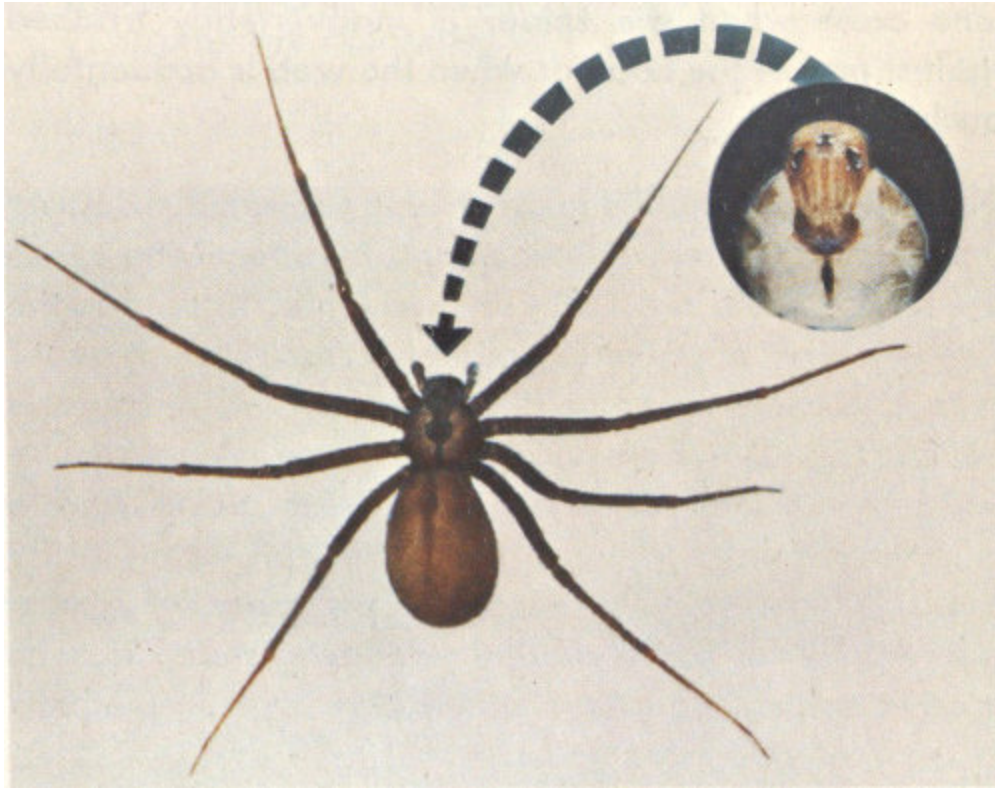
Biology/Behavior. The black widow is generally found in its irregular-shaped web near the ground. Common web sites are under stones, loose bark, or water faucets or in wood piles, rodent burrows, garages, storage buildings, outhouses, and barns. Most human envenomizations occur when the spider is inadvertently trapped against part of the body or when the web is accidentally touched. 5

Avoidance/Control. Frequent cleaning to remove spiders and their webs from buildings and outdoor living areas will decrease the possibility of accidental bites. Routine hose washings of potential spider habitats, such as under steps and around windows and doors, will discourage the black widow from locating in these places. When working in spider-infested areas, wear gloves and a long-sleeved shirt. If warranted, the outside or inside of a building may be sprayed with an approved insecticide. Apply the spray around windows, stairs, closets, or other spider habitats in accordance with instructions on the manufacturer's label.

Envenomization Reaction/Treatment. (See [appendix](#).)

2. BROWN RECLUSE SPIDER

(*Loxosceles reclusa*)



Identification. The brown recluse is a medium-sized spider with a 2- to 4-cm (0.8-1.6 in) legspan and a color range from yellow tan to dark brown. The most distinguishing characteristics are six eyes (most spiders have eight eyes) arranged in a semicircle of three pairs on top of the head, and a violin-shaped marking extending from the area of the eyes to the abdomen.

Distribution. Brown recluse spiders occur throughout an area of the Southcentral States, including Alabama, Arkansas, Georgia, Illinois, Indiana, Iowa, Kansas, Kentucky, Louisiana, Ohio, Oklahoma, Mississippi, Missouri, Tennessee, and Texas. Localized populations of this spider, probably imported from the Southcentral States, have been reported from Arizona, Wyoming, California, Florida, New Jersey, North Carolina, Pennsylvania, and Washington, D.C. Due to the mobility of the United States citizenry, specimens can be easily transported in household goods from the spider's home range in the Southcentral United States to any other area of the country. Under favorable conditions the relocated spiders can survive for an extended period of time and possibly become established.

Biology/Behavior. Within its range, the brown recluse spider will readily establish populations inside parts of buildings which are generally dry, littered, and undisturbed for long periods of time. The spider also can be found outside in protected areas (under rocks and loose bark). Members of this species are nonaggressive and normally attempt to escape whenever they are threatened. Thus, most instances of bites occur when the spider is inadvertently trapped, such as when the victim puts on clothing in which the spider is hiding, steps on a wandering spider at night, or cleans closets or other storage areas where the spider resides.

Avoidance/Control. Any of the following actions will help prevent contact with the brown recluse spider: shake out clothing and bedding before use; eliminate collections of papers and unused boxes; thoroughly clean beneath and behind furniture; remove spiders, webs, and egg cases from living and storage areas; and properly use appropriate insecticides.

Envenomization Reaction/Treatment. (See [appendix](#).)

3. CENTIPEDES

(Class: Chilopoda)



Identification. Centipedes are multisegmented elongate arthropods with a distinct head and one pair of legs, or appendages, per segment. Size is species dependent, with body length ranging from approximately 2.5 to 25 cm (1-10 in) or more. The number of legs can therefore vary from 15 to 100 pairs or more, depending upon the species.

Distribution. The range of centipedes varies with the species; however, these arthropods are most numerous in the southern half of the United States.

Biology/Behavior. Throughout the day, centipedes hide under rocks, boards, or bark and in cracks, crevices, closets, basements, and other moist, protected locations. At night, they come out of hiding to hunt for prey which usually consists of insects and other small arthropods. Centipedes inject venom through two powerful claws located on the ventral side of the body immediately behind the head. Contrary to popular belief, centipedes cannot inject venom through their numerous legs. The majority of centipedes found in the United States are small and not a threat

to man. Human centipede envenomizations occur when a relatively large centipede is accidentally picked up, stepped on, or otherwise trapped against the body.

Avoidance/Control. When camping in a centipede-infested area, carefully invert and shake out sleeping bags, clothes, or other items left in contact with the ground. Always wear shoes when walking about at night, and wear leather gloves when moving rocks or trash from the ground. Usually centipedes are not sufficiently numerous in any one location to warrant chemical control.

Envenomization Reaction/Treatment. (See [appendix](#).)

10

4. CONENOSE OR KISSING BUGS (*Triatoma spp.*)



Identification. Species of the genus *Triatoma* have the elongate (cone-shaped) head which is characteristic of the family Reduviidae. Hence, the name “conenose bugs” is often used to describe these insects. Orange and black markings are usually present where the abdomen extends laterally past the folded wings. These insects are flattened dorsally, and this trait allows them to effectively hide in small cracks and crevices. The size of mature adults varies from approximately 1 to 3 cm (0.4-1.2 in) depending upon the species.

Distribution. *Triatoma* species that attack humans are generally located in the southern half of the United States.

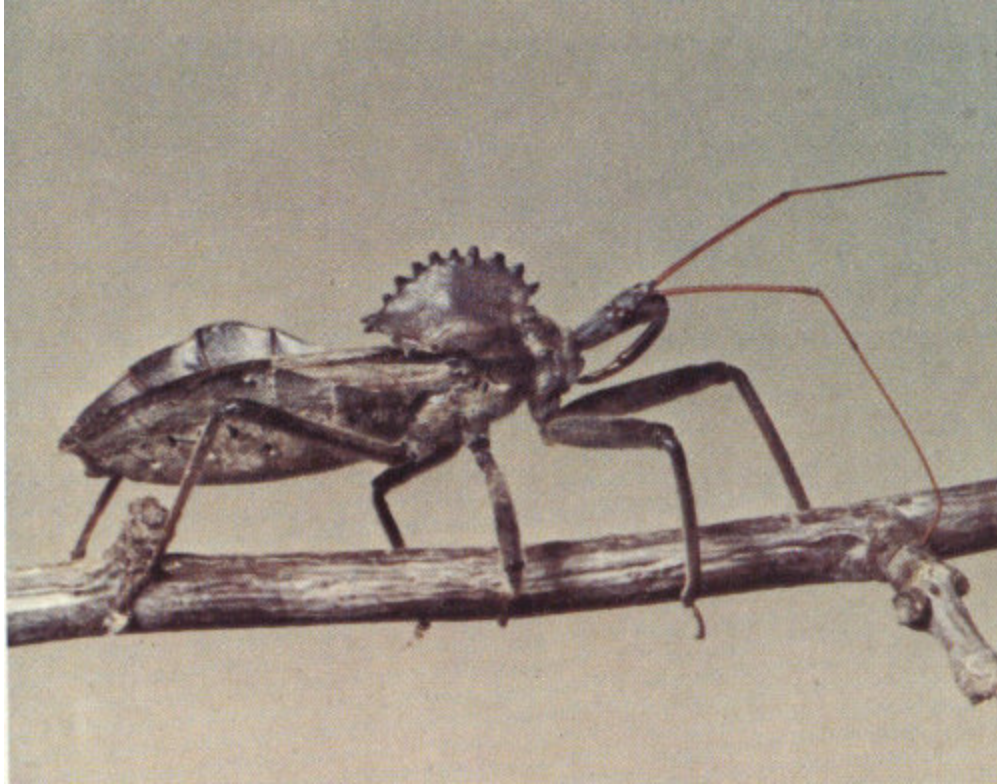
Biology/Behavior. Conenose bugs are nocturnal insects. They take their blood meals at night and hide in any available crack or crevice between feedings. *Triatoma*, as a group, normally feed on small mammals; but in the absence of their preferred hosts, several species will readily feed on humans. They are commonly referred to as “kissing bugs” because their blood meals are occasionally taken from the area around the human lips. Some of the common sites of human attack, in order of frequency, are the hands, arms, feet, head, and trunk. This feeding pattern can be expected since kissing bugs have not been found to feed through clothing.

Avoidance/Control. Since these bugs can feed undetected on a sleeping victim, it is difficult to avoid their attack in heavily infested areas. Exposure to kissing bugs can be minimized, however, by destroying underbrush, rubbish, wood rats’ nests, and any other small-animal habitat located near a residence. When an infestation is detected inside a house, two or three treatments with an appropriate insecticide at 10-day intervals should provide control. To obtain the best results, thoroughly treat all windows, baseboards, walls, cracks, crevices, and bedsprings.

Envenomization Reaction/Treatment. (See [appendix](#).)

5. WHEEL BUG

(*Arilus cristatus*)



Identification. *Arilus cristatus* has the small narrow head characteristic of the Reduviidae family of Hemiptera (true bugs). This insect is mouse gray in color and approximately 2.5 to 4 cm (1-1.6 in) long. A cogwheel-like crest on the dorsal side of the prothorax is distinctive to this insect and accounts for its popular name, “wheel bug.”

Distribution. Wheel bugs are generally found in the southern two-thirds of the United States.

Biology/Behavior. Wheel bugs are predacious on soft-bodied insects. Human envenomizations are usually the result of accidental contact while handling vegetation, boards, or other objects. The bug penetrates the skin with its “beak,” or proboscis, and injects a salivary fluid normally used in killing its insect prey. ³¹

Avoidance/Control. The best way to prevent wheel bug contact is to be able to identify this unusual insect and avoid it. Children should be instructed not to handle it. Wearing leather gloves while working outside

will prevent bites which occur when the wheel bug is accidentally picked up with vegetation or other debris. Since wheel bugs are predacious on many harmful insects and are generally considered beneficial, control is not recommended.

Envenomization Reaction/Treatment. Wheel bug bites are characterized by immediate intense pain which usually subsides in 3 to 6 hours. Since all reported bite reactions have been localized and self-limiting, specific treatment measures are not provided for *A. cristatus* bites. However, anaphylactic shock has resulted from the bite of other Hemiptera and should be considered a possibility with wheel bug bites. ¹²

STINGING ARTHROPODS

1. ANTS

(Order: Hymenoptera; Family: Formicidae)



Fire Ant Mound



Harvester Ant Mound

Identification. Like other Hymenoptera, females of dangerous ant species have a stinger at the posterior tip of the abdomen. Their constricted “waist” between the thorax and posterior abdomen and their elbowed antennae distinguish them from similar insects. The dangerous United States species belong to genera *Solenopsis* (fire ants) or *Pogonomyrmex* (harvester ants). The easiest way to distinguish these ants is by their mounds. Fire ants build elevated earthen mounds 8-90 cm (3-36 in) high, leaving the surrounding vegetation relatively undisturbed. In contrast, harvester ant mounds are usually flat or slightly elevated and are surrounded by a defoliated area 0.6-3 m (2-10 ft) or more in diameter. Also, harvester ants are 2 to 3 times larger than fire ants. 15

Distribution. The Florida harvester ant, *P. badius*, distributed throughout the Southeastern States, is the only harvester ant known to occur east of the Mississippi River. West of the Mississippi, the red harvester ant (*P. barbatus*), the California harvester ant (*P. californicus*), and the western harvester ant (*P. occidentalis*) are found in different individualized distribution areas. The two most important fire ant species were introduced

into the United States from South America. The red imported fire ant (*S. invicta*) was introduced about 1940 and is well established in the Southeastern States. Currently, this ant infests Alabama, Arkansas, Florida, Georgia, Louisiana, Mississippi, North and South Carolina, and Texas. The black imported fire ant (*S. richteri*) was introduced about 1918; its distribution is limited to an area in northeastern Mississippi and northwestern Alabama. Two less important native species, *S. geminata* (tropical or native fire ant) and *S. xyloni* (southern fire ant), occur in localized areas of a few southern states.

Biology/Behavior. Ants of the *Solenopsis* and *Pogonomyrmex* genera are diurnal soil-inhabiting insects that prefer to nest in open areas. These ants are social insects; their colonies consist of at least one reproductive queen, several males, and many nonreproductive female workers. Fire and harvester ants are particularly dangerous because workers defending their nest can inflict repeated stings, and multiple human envenomizations may occur when a victim comes in contact with a mound. 16

Avoidance/Control. Familiarization with the appearance of the ant mounds is necessary in order to avoid them and prevent accidental envenomizations. Since small children are often the victims of multiple stings, parents should insure that the mounds of dangerous ant species are not present in play areas. This is particularly important where the red imported fire ant occurs. Mound treatments with mirex have been effective in controlling *Solenopsis* and *Pogonomyrmex* ants. The Environmental Protection Agency limits the use of this insecticide, and it must be applied in strict accordance with EPA and product label instructions.

Envenomization Reaction/Treatment. (See [appendix](#).)

17

2. BEES

(Order: Hymenoptera; Family: Apidae)



Identification. Two families of bees are commonly associated with human envenomization: Apidae (honey bees and bumble bees) and Halictidae (sweat bees). Only the honey bees and bumble bees are considered dangerous. Both are robust hairy insects with two pairs of membranous wings which they hold at an angular position over the body when at rest. Microscopic examination shows that bee thoracic hairs are individually “feathered.” In contrast, wasp thoracic hairs are smooth.

Distribution. Dangerous species of bees are distributed throughout the United States.

Biology/Behavior. 1. Honey Bees. Since these insects are not native to this continent, north of Mexico, all original honey bee colonies in the United States were imported. *Apis mellifera*, the cosmopolitan bee of commerce, is responsible for most bee stings and many deaths.^[10] This species is^[18] separated into several races, with the gold Italian and the black or gray^[5] Caucasian races making up the majority of the bees found in this country.

The aggressive Brazilian (hybrid African) bee is currently restricted to the South American continent. Honey bees are highly social insects. Their colony consists of an egg-laying queen, drones to fertilize the queen, and workers to gather food and care for the young. Honey bees can release “alarm odors” associated with the sting gland, which cause other bees in the vicinity of an odor-marked victim to attack and inflict multiple stings. Unlike other hymenopterous insects, the honey bee worker has a barbed stinger and can sting only once; to escape, the bee must leave its entire stinging apparatus attached to the skin of its victim. As important pollinators of crops, honey bees are very beneficial to man.

2. Bumble Bees. The bumble bees are considered to be the most primitive of the social bees. Their colonies lack much of the structure and highly evolved behavior of the honey bees; however, like the honey bees, the bumble bees are diurnal plant feeders and important pollinators of crops. A typical colony consists of at least one queen, several males, and numerous workers. Only young fertilized queens survive the winter to establish new colonies the next spring. The nests are normally located deep in undisturbed ground, like fence rows, and are supplied with a mixture of pollen and honey. During late summer, a colony usually contains between 100 and 500 bees. Although bumble bees are 2 to 3 times larger than honey bees, they are neither as aggressive nor as abundant as the honey bees, and therefore not as dangerous.

Avoidance/Control. The best way to avoid Hymenoptera stings is to prevent human contact with these insects. This is especially important for persons hypersensitive to bee venom. Some preventive steps are:

1 Avoid outdoor activities in unfamiliar areas where stinging insects are known to occur. 19

2 When outdoors, don’t use floral-scented cosmetic products or leave sweet beverages or foods exposed in areas where they might attract bees.

3 Avoid garbage collection areas, which attract Hymenoptera.

4 When outdoors, always wear shoes and, if possible, a long-sleeved shirt, long pants, or other protective clothing. Don’t wear clothing with a

bright floral print or loose-fitting clothing in which stinging insects may be trapped.

5 Don't make rapid movements around stinging insects or intentionally disturb either the insects or their nests.

6 Eliminate all Hymenoptera nests around inhabited areas (wild honey bee colonies may be removed by a local beekeeper). When necessary, an insecticide such as 5% carbaryl dust may be applied in a nesting area within a building to eliminate the nuisance colony.

7 Educate young children on the hazards of venomous arthropods.

Envenomization Reaction/Treatment. (See [appendix](#).)

20

3. SCORPIONS

(Class: Arachnida; Order: Scorpionida)



Identification. Scorpions are crablike in appearance, with pincers attached to their two front appendages. They also possess a five-segmented “tail” terminating in a bulbous structure with a prominent stinger. The body size of mature scorpions varies from 2 to 10 cm (0.8-4 in) or more, depending upon the species.

Distribution. Scorpions are most prevalent in warmer climates. In the United States the two most dangerous scorpion species, *Centruroides sculpturatus* and *C. gertschi*, have been found only in Arizona.

Biology/Behavior. During daylight hours scorpions hide under objects or debris on the ground. At night, they come out of hiding to search for food, which consists primarily of ground-inhabiting arthropods. The prey is grasped with the pincers and stung by rapidly bringing the stinger forward directly over the head. This quick stinging motion is also used for defense, as is usually the case with human envenomizations. 21

Avoidance/Control. To avoid stings in scorpion-infested areas: remove accumulations of boards, rocks, and other debris; wear leather gloves to remove any item from the ground; inspect and shake out clothing or shoes before donning them; and when moving about at night, turn on lights to avoid contacting these nocturnally active arthropods. Insecticides that are commonly used for household cockroach control are also useful for controlling scorpions. Special care should be taken to treat around all baseboards thoroughly.

Envenomization Reaction/Treatment. (See [appendix](#).)

22

4. VELVET ANTS

(Order: Hymenoptera; Family: Mutillidae)



Identification. Female velvet ants, wingless and antlike in appearance, are actually wasps that only resemble ants. The males are winged and usually larger than the females. Both male and female are covered with a velvety pubescence which is normally bright red, orange, or yellow. The female has a formidable stinger at the posterior end of her abdomen which she can use to inflict a painful envenomization. The length of a mature mutillid wasp varies from 1 to 2.5 cm (0.4-1 in). In some localities, mutillids are commonly referred to as cow killers, mule killers, or wooly ants.

Distribution. The Mutillidae is a large family with most species distributed in the southern and western parts of the United States.² However, one species, *Dasymutilla occidentalis*, is commonly found on the sandy beaches of Lake Erie and during the summer months causes barefoot bathers much distress.⁵

Biology/Behavior. Female velvet ants are solitary, diurnal, parasitic wasps. An efficient stinging apparatus and an extremely thick

exoskeleton allow the female mutillid to conspicuously, yet safely, run about in the open searching for a suitable place to lay her eggs. Most cases of human envenomization occur when the female wasp is accidentally touched while roaming about on the ground or is trapped against the body in clothing or bedding.

Avoidance/Control. The solitary and roaming nature of velvet ants makes it very difficult to predict or designate ways to avoid contacting them. In areas like the beaches of Lake Erie where mutillids are known to frequent, however, potential victims should be made aware of the velvet ant hazard and how to avoid accidental envenomization. Since mutillids are not gregarious and do not congregate in accessible places, chemical control is not feasible.

Envenomization Reaction/Treatment. (See [appendix](#).)

24

5. WASPS

(Order: Hymenoptera; Family: Vespidae)



Bald-Faced Hornet



Yellow Jacket

Identification. This information applies to all dangerous wasps found in the United States except the mutillids (previously discussed). Wasps are elongate insects with three distinct body parts and four wings. As with all winged forms of Hymenoptera, the front pair of wings is larger than the hind pair. Only the female wasps are equipped with a stinger at the posterior end of the abdomen. To distinguish wasps from similar-appearing bees, the specimen's thoracic hairs should be examined microscopically to determine if they are smooth (a wasp) or feathered (a bee). In the United States, members of the paper-wasp family (Vespidae) are generally considered to be the most dangerous because of their social nesting habits, aggressive behavior, and abundance. Two representative members of the Vespidae family found in North America are the bald-faced hornet, *Vespula maculata*, and a common yellow jacket, *V. pennsylvanica*. Although it takes an expert to identify most wasps according to species, the paperlike nests of the vespids make their identification easy. Another important group of

dangerous wasps easily distinguished by their nests are the mud daubers. These wasps usually build their mud nests in the corners of man-made structures. Besides their distinctive nests, mud daubers are identified by their dark shiny color and long narrow waist.

Distribution. Dangerous species of wasps can be found throughout the United States.

Biology/Behavior. Most vespids are social insects that feed primarily on other arthropods. These wasps build paperlike nests by masticating wood fiber into pulp. In addition to the social wasps, many types of solitary wasps are capable of envenomization; but these wasps are easy to avoid since they are neither abundant nor aggressive. Female members of the dangerous wasp species will readily attack and inject their venom if someone disturbs or annoys them. Wasp stingers do not become detached, and a single wasp can sting a victim several times before retreating. Social wasps are particularly dangerous because a disturbance of their nest may result 26 in attack by numerous wasps.

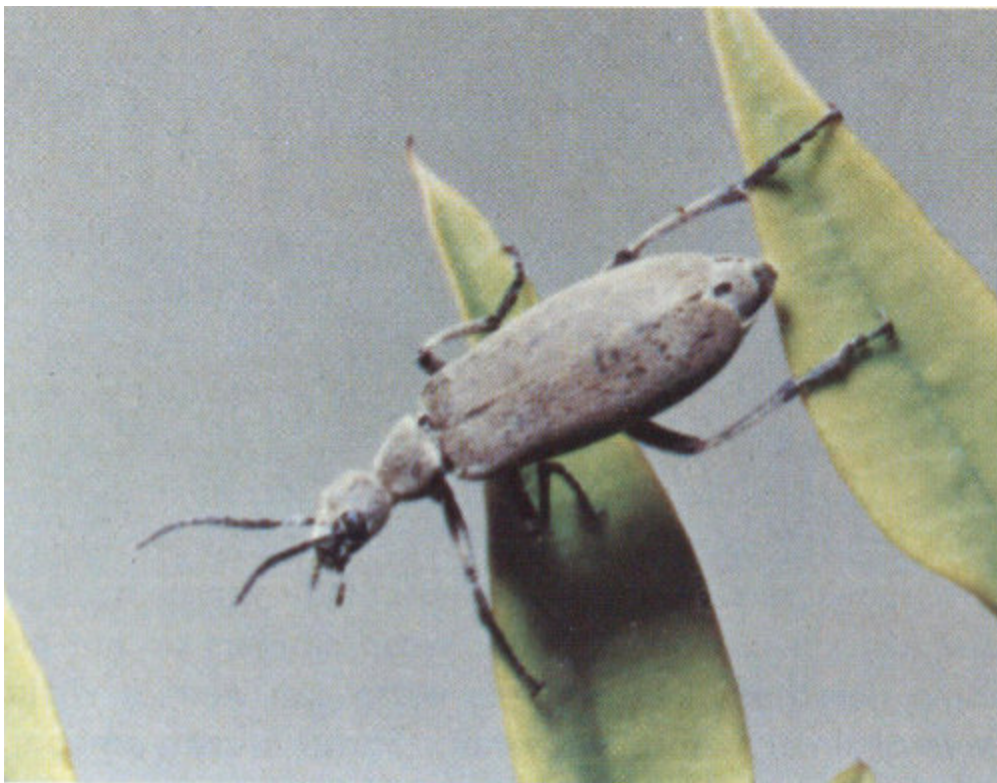
Avoidance/Control. Avoid garbage accumulation sites, which attract wasps. To avoid wasp stings, recognizing the nests of the dangerous species in the local area, is important. If possible, remove and destroy all nests found around inhabited areas. Wasps can be controlled by treating their nests with insecticide—preferably at night, when the wasps are least active. Treat nests found above the ground with a commercial wasp spray or other insecticide recommended by an entomologist. To treat wasp nests in the 38 ground or in houses, a 5% carbaryl dust may be used.

Envenomization Reaction/Treatment. (See [appendix](#).)

URTICATING/VESICATING ARTHROPODS

1. BLISTER BEETLES

(Order: Coleoptera; Family: Meloidae)



Identification. The Meloidae are narrow elongate beetles characterized by a “neck” (pronotum) which is distinctly narrower than its head or wings. Adult beetles range in body length from 1 to 2 cm (0.4-0.8 in) and vary considerably in their coloration. 27

Distribution. In general, blister beetles are found in greater numbers in the western half of the United States. 5

Biology/Behavior. The immature stages of the blister beetle are usually predacious on other insects and are not harmful to man. Adult blister beetles release a clear amber fluid by rupture of thin membranes in the leg joints or other segmented areas of the body. The release of this fluid, which contains a vesicating agent called cantharidin, is triggered by pressure against the body of the beetle. Light pressure exerted by clothing or by brushing off a beetle is usually sufficient to cause the release of its vesicating fluid. The adult blister beetles are readily attracted to bright white light, and many cases of human exposure occur at night around such lights. Since the adults are plant feeders, some cases of human vesication occur as persons move through vegetation infested with blister beetles.

Avoidance/Control. In areas with an abundance of blister beetles, use yellow light bulbs for outdoor lighting. If a meloid beetle lands on the skin, blow it off, do not crush it. Since cantharidin is distributed throughout the beetle's body, crushing the beetle against exposed skin would result in maximum cantharidin exposure. Dermatoses resulting from blister beetle contact are seasonal, with the greatest number of vesicating incidents in the United States occurring in July, August, and September. Chemical control usually is not recommended because of the mobility and wide distribution of blister beetles.

Envenomization Reaction/Treatment. This information is not included in the appendix because the envenomization reaction generally consists of a superficial linear bulla which does not require emergency treatment.

2. URTICATING CATERPILLARS

a. IO MOTH CATERPILLAR (*Automeris io*)



Identification. The *Automeris io* caterpillar is the larva of the Io moth, which belongs to the moth family Saturniidae. A full-grown caterpillar is about 5 to 8 cm (2-3 in) long, pale green, with lateral strips of red or maroon over white running the length of the body. Near the center of each body segment is a partial row of tubercles armed with radiating green and black spines. Many of these spines are venomous, and their tips are connected to rather large individualized poison glands.

Distribution. In the United States, the Io moth is found in the states east of the Rocky Mountains. ⁶⁸

Biology/Behavior. Io moth larvae feed on the leaves of a variety of ²⁹ plants, including corn and willow. ¹⁷ In most areas they produce only one annual generation, emerging as a moth in the spring or summer and overwintering as a pupa. In south Texas, however, two generations occur, with one developing in May-July and the second in October-December. Therefore, the urticating larval stages can be found anytime from early

spring to late fall, depending upon the area of the country and climatic conditions. As with most urticating caterpillars, *A. io* envenomizations normally occur when the victim inadvertently contacts caterpillar-infested vegetation. At the instant the victim's skin touches this caterpillar, the spine tips break off in the skin, thus allowing toxin to flow out of the hollow spines and onto the skin.

Avoidance/Control. Wear gloves, a long-sleeved shirt, and long pants when working in an infested area. Children should be warned not to handle caterpillars. If necessary, infested vegetation may be treated with an appropriate insecticide.

Envenomization Reaction/Treatment. (See [appendix](#).)

30

b. PUSS CATERPILLAR
(*Megalopyge opercularis*)



Identification. The puss caterpillar is the larva of a lepidopterous group commonly referred to as “flannel moths.” In some parts of the United States the larva is incorrectly referred to as an “asp.” When fully mature, the larvae are near white to dark gray in color, 2 to 3 cm (0.8-1.2 in) long, and completely covered dorsolaterally with hairs that cause them to resemble elongate tufts of cotton. Some of the hairs are venomous, and when they penetrate the skin a “toxin” passes from an underlying gland through the hairs at the points of contact. The color variation of the larvae in this species is dependent upon larval age, locality, and time of year. In Texas, where this species is most abundant, the spring/summer generation of caterpillars is usually lighter in color than the fall generation.

Distribution. This species has been recorded primarily from the 31 Southeastern States, including Alabama, Arkansas, Florida, Georgia, Louisiana, Maryland, Mississippi, Missouri, North Carolina, South Carolina, Texas, and Virginia.

Biology/Behavior. In most of the southern area of its range, the puss caterpillar is thought to have two generations per year. The first generation develops in the spring and early summer, while the second generation develops in the fall. After emerging from a cocoon and mating, the female moth lays her eggs on a suitable host plant. In a few days the eggs hatch into larvae, which develop by feeding on the leaves of a wide range of trees and shrubs. The natural enemies of the puss caterpillar usually keep its numbers under control; however, every 4 or 5 years the caterpillars become more numerous, and the number of envenomizations associated with them 66 increases.

Avoidance/Control. Always wear protective clothing such as gloves and a long-sleeved shirt when working in areas heavily infested with puss caterpillars. During periods of *M. opercularis* abundance, children should be instructed to stay away from infested trees and shrubs and not to handle caterpillars. If necessary, heavy puss caterpillar infestations may be treated with an appropriate insecticide.

Envenomization Reaction/Treatment. (See [appendix](#).)

c. SADDLEBACK CATERPILLAR
(*Sibine stimulea*)

32



Identification. The saddleback caterpillar is easy to recognize since its brown sluglike body is covered mid-dorsally with markings that resemble a brown or purplish saddle sitting on a green and white saddle blanket. Upon close examination, stout spines can be observed along the caterpillar's lateral body margin and on its four tubercles. Many of these spines are hairs that are connected at their bases with individual poison glands. Just prior to pupation and subsequent development into a moth, the caterpillar is 2-3 cm (0.8-1.2 in) long.

Distribution. In general, the saddleback caterpillar in the United States is distributed southeast of a diagonal line drawn from Massachusetts through the middle of Texas. ⁶⁸

Biology/Behavior. From May to November, *Sibine stimulea* caterpillars may be found feeding on the leaves of a large variety of trees, shrubs, and other plants. Envenomization usually occurs when the victim accidentally contacts vegetation infested with these caterpillars. At the instant the victim's skin contacts the caterpillar, the spine tips break off, thus allowing toxin to flow out of the hollow spines and into or onto the skin. 33

Avoidance/Control. When working in infested areas, wear gloves, a long-sleeved shirt, and long pants to prevent accidental envenomization. During periods of local heavy infestations, children should be instructed to avoid trees, shrubs, and other vegetation commonly infested with these caterpillars. Local entomologists may be contacted to obtain information on the most effective insecticide for saddleback caterpillar control in a given area.

Envenomization Reaction/Treatment. (See [appendix](#).)

BIBLIOGRAPHY

General

1. Barnard, J.H. Cutaneous responses to insects. JAMA 196:3 (1966).
2. Borror, D.J., and D.M. DeLong. An introduction to the study of insects, 2d ed. New York: Holt, Rinehart and Winston, 1964.
3. Dodge, N.N. Poisonous dwellers of the desert. Popular Series No. 3, Southwestern Monuments Assoc., Box 1562, Globe, Ariz., 1955.
4. Frazier, C.A. Diagnosis and treatment of insect bites. Clin Symp 20:75-100 (1968).
5. Frazier, C.A. Insect allergy: Allergic reactions to bites of insects and other arthropods. St. Louis: Warren H. Green, Inc., 1969.
6. Frazier, C.A. Treatment guide: Common biting and stinging insects and other arthropods. West Point, Pa.: Merck Sharp & Dohme, 1972.
7. Frazier, C.A. Cutaneous manifestations of insect allergy. Cutis 13:1038-1047 (1974).
8. Horen, W.P. Insect and scorpion sting. JAMA 221:894-898 (1972)
9. Infections and bites: Parasites, arthropods and animals, Section 18. *In* D.J. Demis et al. (eds.). Clinical dermatology, vol. 4. Hagerstown, Md.: Harper & Row, 1974.
10. James, M.T., and R.F. Harwood. Herm's medical entomology, 6th ed. London: Collier-MacMillan, 1970.
11. Lyon, J.B. Insect bites and stings. Practitioner 200: 670-677 (1968).

12. Micks, D.W. Insects and other arthropods of medical importance in Texas. *Tex Rep Biol Med* 18:624-635 (1960)
13. Millikan, L.E., and R.S. Berger. Clinical and laboratory diagnosis of arthropod bites. *CRC Crit Rev Clin Lab Sci* 5:201-225 (1974).
14. Parrish, H.M. Deaths from bites and stings of venomous animals and insects in the United States. *Arch Intern Med* 104:198-207 (1959). 35
15. Parrish, H.M. Analysis of 460 fatalities from venomous animals in the United States. *Am J Med Sci* 245:129-141 (1963).
16. Perlman, F. Insects as allergen injectants: Severe reactions to bites and stings of arthropods. *Calif Med* 96:1-10 (1962).
17. Riley, W.A., and O.A. Johannsen. *Medical entomology*, 2d ed. New York: McGraw-Hill, 1938.
18. Russell, F.E. Injuries by venomous animals. *Am J Nurs* 66:1322-1326 (1966).
19. Scott, H.G. Envenomization. DHEW CDC Publ No. (HSM) 72-8121, 1972.

Blister Beetles

20. Lehmann, C.F., et al. Blister beetle dermatosis. *Arch Dermatol* 71:36-38 (1955).
21. Rosin, R.D. Cantharides intoxication. *Br Med J* 4:33 (1967)

Centipedes

22. Cloudsley-Thompson, J.L. *Spiders, scorpions, centipedes and mites*. New York: Pergamon, 1958.

23. Pineda, E.V. A fatal case of centipede bite. J Philipp Med Assoc 3:59-61 (1923).

24. Remington, C.L. The bite and habits of a giant centipede (*Scolopendra subspinipes*) in the Philippine Islands. Am J Trop Med 30:453-455 (1950).

Hemiptera

25. Hall, M.C. Lesions due to the bite of the wheel-bug, *Arilus cristatus* (Hemiptera, Reduviidae). Arch Intern Med 33:513-515 (1924).

26. Marks, M.B. Stinging insects: Allergy implications. Pediatr Clin North Am 16:177-191 (1969).

27. Nichols, N., and T.W. Green. Allergic reactions to “kissing bug” bites. Calif Med 98:267-268 (1963). 36

28. Parsons, D.J. Bedbug bite anaphylaxis misinterpreted as coronary occlusion. Ohio Med J 51:669 (1955).

29. Readio, P.A. Studies on the biology of the Reduviidae of America north of Mexico. Univ Kans Sci Bull 17:5 (1927).

30. Shields, T.L., and E.N. Walsh. “Kissing bug” bite. Arch Dermatol 74:14-21 (1956).

31. Smith, F.D., et al. Insect bite by *Arilus cristatus*, a North American Reduviid. Arch Dermatol 77:324-330 (1958).

32. Wray, D.L. Two cases of Hemiptera biting humans (Hemiptera, Reduviidae). Entomol News 82:228 (1971).

Hymenoptera

33. Barnard, J.H. Studies of 400 hymenoptera sting deaths in the United States. J Allergy Clin Dermatol 52:259-264 (1973).

34. Barr, S.E. Allergy to hymenoptera stings. JAMA 228:718-720 (1974).
35. Busse, W., et al. Protection following honeybee venom immunotherapy in a case of bee sting anaphylaxis. (Abstract) J Allergy Clin Immunol 53:104 (1974).
36. Busse, W.S., et al. Immunotherapy in bee-sting anaphylaxis: Use of honeybee venom. JAMA 231:1154-1156 (1975).
37. Charavejasarn, C.C., et al. Stinging insect hypersensitivity. II. Studies of bee venom antigenicity. (Abstract) J Allergy Clin Immunol 53:104 (1974).
38. Controlling wasps. U.S. Department of Agriculture, Home and Garden Bull No. 122, 1972.
39. Favorite, F.G. The imported fire ant. Public Health Rep 73:445-448 (1958).
40. Insect-sting allergy: Questionnaire study of 2,606 cases. JAMA 193:109-114 (1965).
41. Lichtenstein, L.M., et al. Treatment of honeybee anaphylactic sensitivity with honeybee venom: A case report. (Abstract) J Allergy Clin Immunol 53:104-105 (1974). 37
42. Majeski, J.A., et al. Acute systemic anaphylaxis associated with an ant sting. South Med J 67:365-366 (1974).
43. McCormick, W.F. Fatal anaphylactic reactions to wasp stings. Am J Clin Pathol 39:485-491 (1963).
44. McLean, J.A. Management of insect sting reactions. Mod Treat 5:814-824 (1968).
45. Morse, R.A., et al. Protective measures against stinging insects. NY State J Med 59:1546-1548 (1959).
46. Reisman, R.E. Stinging insect allergy—treatment failures. J Allergy Clin Immunol 52:257-258 (1973).

47. Spencer, J.T. Treatment of yellow jacket stings. JAMA 206:649 (1968).
48. Torsney, P.J. Treatment failure: Insect desensitization. J Allergy Clin Dermatol 52:303-306 (1973).
49. Valentine, M.D., et al. Blocking antibody to bee venom: Induction by venom and not by whole body extracts. (Abstract) J Allergy Clin Immunol 53:105 (1974).

Scorpions

50. Masco, H.L. Scorpion bite treatment with chlorpromazine. JAMA 212:2122 (1970).
51. Stahnke, H.L. Scorpions. Tempe, Ariz.: Arizona State College Bookstore, 1949.
52. Vachon, M. The biology of scorpions. Endeavor April:80-89 (1953).

Spiders

53. Anderson, P.C. What's new in loxoscelism? Mo Med 70:711, 712, 718 (1973).
54. Atkins, J.A., et al. Probable cause of necrotic spider bite in the Midwest. Science 126:73 (1957).
55. Auer, A.I., and F.B. Hershey. Surgery for necrotic bites of the brown spider. Arch Surg 108:612-618 (1974).
56. Cheng, A.K. USAF School of Aerospace Medicine, San Antonio, Tex. Personal communication, 1974. 38
57. Gladney, W.J. Controlling the brown recluse spider. U.S. Department of Agriculture Leaflet No. 556, 1972.

58. Hammon, P.J., and W.H. Newton. The brown recluse and the black widow spiders. Texas A&M Univ, Fact Sheet L-623 (1970).
59. Hershey, R.B., and C.E. Aulenbacher. Surgical treatment of brown spider bites. Ann Surg 170:300-308 (1969)
60. Horen, W.P. Arachnidism. Clin Med (Aug):41-43 (1966)
61. Russell, F. E., et al. Bites of spiders and other arthropods, Section 16, pp. 865-867. In H.F. Conn (ed.). Current therapy. Philadelphia: W.B. Saunders, 1974.
62. Taylor, E.H., and W.F. Denney. Hemolysis, renal failure and sudden death, presumed secondary to bite of brown recluse spider. South Med J 59:1209-1211 (1966).
63. The brown recluse. Public Health Service Publ No. 2062, 1970.

Urticating Caterpillars

64. Daly, J.A., and B.L. Derrick. Puss caterpillar sting in Arkansas. South Med J 68:893-894 (1975).
65. Kendall, R.O. San Antonio, Tex. Personal communication, 1975.
66. McGovern, J.P., et al. *Megalopyge opercularis*: Observations of its life history, natural history of its sting in man, and report of an epidemic. JAMA 175: 121-124 (1961).
67. Micks, D.S. Clinical effects of the sting of the “puss caterpillar” (*Megalopyge opercularis* S. & A.) on man. Tex Rep Biol Med 10:399-405 (1952).
68. Mitchell, R.T., and H.S. Zim. Butterflies and moths. New York: Golden Press, 1964.

APPENDIX: ARTHROPOD ENVENOMIZATION REACTIONS AND TREATMENT

The material presented in this appendix was prepared by Lt Col John C. Moseley, USAF, MC, Dermatologist, Wilford Hall USAF Medical Center, Lackland AFB, Texas.

BLACK WIDOW SPIDER

REACTIONS

IMMEDIATE: Pinprick sensation from the bite

LOCAL: Dull numbing pain; two red puncture marks at bite site. Pain peaks at 1-3 hours and persists 12-48 hours

REGIONAL:

1. Severe muscular pain and spasm
2. Rigid boardlike abdomen
3. Tightness in chest and pain on inspiration

GENERAL:

1. Rigidity and spasm of all large muscle groups; excruciating abdominal pain
2. Convulsions, paralysis
3. Shock

DEATHS: 4-5% of untreated cases due to neurotoxic effect of the venom

TREATMENT

LOCAL FIRST AID: Ineffective and unnecessary

SYSTEMIC:

1. Antivenin—1 vial IM (carefully follow package insert instructions) after testing for horse-serum sensitivity
2. Calcium gluconate—10 ml of 10% solution given IV immediately and prn to control muscle pain
3. Muscle relaxants—Give continuously over 8-16-hr period to relieve intensity of muscle spasm and pain
4. Treat for shock as necessary ([Chart 2](#))

40

BROWN RECLUSE SPIDER

REACTIONS

IMMEDIATE: Very little pain, if any

LOCAL:

1. 2-8 Hours—Mild to severe pain with redness and vesiculation at bite site, followed by ischemia
2. 3-4 Days—Star-shaped firm area of deep-purple color with necrosis
3. 7-14 Days—Central area of depression and ulceration
4. 21 Days—Healing and scar formation. May not heal sufficiently and may require skin grafting

SYSTEMIC:

1. Fever, chills, nausea, vomiting, weakness, joint pain
2. Morbilliform or petechial generalized eruption
3. Occasionally serious hematological disturbances—hemolytic anemia, thrombocytopenia

DEATHS: Reported; usually in children, due to renal failure and hematological abnormalities.

TREATMENT

LOCAL FIRST AID: None effective or necessary

LOCAL: Within 4 hours of the bite, locally excise bite site

SYSTEMIC:

1. Systemic corticosteroids is the treatment of choice and must be administered immediately—methylprednisolone (Depo-Medrol) 80 mg IM, followed by prednisone 60 mg a day for 3 days and gradually tapered over a 10-14-day course of therapy
2. Heparin therapy may reduce the disseminated intravascular coagulation phenomenon
3. Analgesics for pain

41

CENTIPEDES

REACTIONS

IMMEDIATE: Pain, often intense

LOCAL: 1-4 hours—A two-puncture wound at the site with redness, swelling, and a burning aching pain that subsides in 4-5 hours

REGIONAL: Rare—Purpura of an entire limb

ANAPHYLAXIS AND DEATH: None reported in the United States

TREATMENT

LOCAL FIRST AID:

1. Wash with soap and water
2. Apply ammonia in 10% solution
3. Apply cool wet dressings of a saturated magnesium sulfate solution

SYSTEMIC: Analgesics for pain

42

TRUE BUGS (KISSING BUGS)

REACTIONS

IMMEDIATE: Usually no sensation, occasionally mild pain

LOCAL: (Four distinct reactions depending on degree of sensitivity):

1. Papule with a central punctum
2. Small vesicles grouped around bite site with swelling and little redness
3. Giant urticarial lesion with central punctum and surrounding brawny edema
4. Hemorrhagic nodular-to-bullous lesions on hands and feet—the characteristic “kissing bug bites”

ANAPHYLACTIC SHOCK: Rare, but reported

DEATHS: None reported

TREATMENT

LOCAL FIRST AID: Wash with soap and water

SYSTEMIC: Oral antihistamines ([Chart 1](#))

43

ANTS

REACTIONS

IMMEDIATE: Fierce burning pain lasting minutes

LOCAL: (Fire Ants)

Minutes—Wheal formation

2-4 Hours—Clear fluid-filled vesicles

8-10 Hours—Cloudy fluid-filled vesicles

12-24 Hours—Umbilicated pustules on a red base, with pain and tenderness

3-8 Days—Lesions resolve; may leave scars

SYSTEMIC: Due to allergic sensitization; severity and speed of onset related to degree of sensitivity. May see wheezing, urticaria, abdominal cramps, generalized edema, nausea, vomiting, dizziness, confusion, shock

ANAPHYLAXIS AND DEATH: Rare, but reported

TREATMENT

LOCAL FIRST AID:

1. Wash sites with soap and water
2. Apply ice packs or cold compresses
3. Apply baking soda and water paste

SYSTEMIC:

1. Epinephrine (1:1,000) subcutaneous injection (0.2-0.5 ml in adults and 0.1-0.3 in children). Repeat in 5-10 min if necessary
2. Antihistamine (Benadryl, 50 mg IM)—Adult
3. Oral antihistamines ([Chart 1](#))
4. Treat for shock as necessary ([Chart 2](#))

BEEES AND WASPS

REACTIONS

IMMEDIATE: Pain

LOCAL: Within 1-4 hours—Appearance and subsidence of wheal and red flare; may see intense local swelling in region of sting

SYSTEMIC:

1. Mild reaction—Generalized urticaria, itching, malaise, anxiety
2. Moderate reaction—Any of the above plus generalized edema, tightness in the chest, wheezing, abdominal pain, nausea, vomiting, dizziness
3. Severe reaction—Any of the above plus labored breathing, difficulty in swallowing, hoarseness or thickened speech, marked weakness, confusion, feeling of impending disaster
4. Shock—Cyanosis, fall in BP, collapse, incontinence, unconsciousness
5. Delayed serum-sickness-like reaction (10-14 days after sting)—Fever, lymphadenopathy, malaise, headache, urticaria, polyarthritis

ANAPHYLAXIS AND SUDDEN DEATH: Many cases reported, usually in adults

TREATMENT**LOCAL FIRST AID:**

1. Remove bee stinger from sting site by gently scraping with fingernail or blade to prevent further venom injection from attached venom sac
2. Wash site with soap and water
3. Apply ice packs or ammonia in 10% solution
4. Apply baking soda and water paste
5. Elevate and rest involved limb

SYSTEMIC:

45

1. Epinephrine (1:1,000) subcutaneous injection (0.2-0.5 ml in adults and 0.1-0.3 ml in children). Repeat in 5-10 min if necessary
2. Analgesics (ASA, Tylenol) for pain
3. Antihistamines ([Chart 1](#))—Useful only for urticarial and pruritic reactions
4. Treat for shock as necessary ([Chart 2](#))

LONG-TERM MANAGEMENT: (for hypersensitive patients)

1. Medic-alert tag or bracelet

2. Emergency treatment kit and instructions for use
3. Program for desensitization

SCORPIONS

REACTIONS

IMMEDIATE: Severe sharp pain

LOCAL:

1. Dangerous neurotoxic species—Pins-and-needles sensation with no local swelling or discoloration (found in Arizona only)
2. Comparatively harmless species (not neurotoxic)—Local swelling and discoloration at sting site

SYSTEMIC (neurotoxic species only—within 1-3 hours):

1. Hypoesthesia and numbness or drowsiness
2. Itching of nose and throat
3. Impaired speech and tightness of jaw muscles
4. Restlessness and muscle twitching
5. Muscle spasms with pain, nausea, vomiting, incontinence, convulsions
6. Respiratory and/or circulatory distress

ANAPHYLAXIS: Rare, but reported with non-neurotoxic species

DEATHS: Occasional; due to neurotoxic species

TREATMENT

LOCAL FIRST AID:

1. Apply ice packs
2. Apply tourniquet if possible and as near sting site as possible. Loosen briefly every 10-15 minutes

WARNING: Do *not* use morphine or opiates since they increase toxic effects

SYSTEMIC:

1. Specific antivenin available for many dangerous species; administered early, may be lifesaving
2. Calcium gluconate—10 ml of 10% solution IV immediately and prn to control muscle pain
3. Phenobarbital—30-60 mg orally for sedation and control of convulsions
4. Treat for shock as necessary ([Chart 1](#))

47

URTICATING CATERPILLARS

REACTIONS

IMMEDIATE: Severe burning pain

LOCAL:

1. Numbness and swelling of area inflicted with severe radiating pain
2. Possible double row of parallel red punctuate marks forming a gridlike tract along the path of the caterpillar
3. Swelling of regional lymph nodes
4. Late foreign-body reaction to unremoved spines

SYSTEMIC:

1. Nausea, vomiting, fever
2. Headaches
3. Shock and convulsions (rare)

DEATHS: None reported

TREATMENT

LOCAL FIRST AID:

1. Repeated stripping using adhesive or cellophane tape to remove spines
2. Apply ice packs
3. Apply baking soda and water paste

SYSTEMIC:

1. For severe pain give meperidine hydrochloride (Demerol, 50-100 mg PO or IM), morphine sulfate (0.25 subcutaneous), codeine phosphate (0.5 g PO)

NOTE: Aspirin is generally not effective

2. Shock ([Chart 2](#))

48

CHART 1
ANTIHISTAMINES

GROUP	TRADE NAME	AVERAGE ORAL BASE		SEDATION
		ADULT	CHILD	
GENERIC NAMES				
Ethanolamines				
diphenhydramine•HCl	Benadryl	50 mg q 4-6h	25 mg q 4-6h	++++
diphenhydramine theophyllinate	Dramamine	50 mg q 4h	25 mg q 4h	++++
Ethylenediamine				
tripelennamine	Pyribenzamine	50 mg q 4-6h	25 mg q 4-6h	+++
Alkylamines				
chlorpheniramine maleate	Chlor- Trimeton	4 mg q 6h	2 mg q 6h	++
brompheniramine maleate	Dimetane	8 mg q 6h	4 mg q 6h	+

triprolidine•HCl	Actidil	2.5 mg q 8h	1.25 mg q 8h	++
Cyclizines				
hydroxyzine•HCl	Atarax	25-100 mg q 6h	10-25 mg q 6h	+
Miscellaneous				
ciproheptadine•HCl	Periactin	4 mg q 6h	2 mg q 6h	+++
promethazine	Phenergan	25-50 mg q 6-8h	12.5-25 mg q 6-8h	++++

Chart 2 Treatment of Anaphylaxis—Shock

IMMEDIATE TREATMENT	MILD REACTION TREATMENT	SEVERE REACTION TREATMENT
REACTION: Conjunctivitis, Rhinitis, Urticaria, Pruritus, Erythema		
Epinephrine•HCl 0.3 ml (1:1,000) IM	Diphenhydramine•HCl 50 mg PO q 6h	
Diphenhydramine•HCl 50 mg PO		
REACTION: Laryngeal edema		
Epinephrine•HCl 0.3 ml (1:1,000) IM	Diphenhydramine•HCl 50 mg q 6h IM or PO	Oxygen
Diphenhydramine•HCl 50 mg IV	Ephedrine sulfate 25 mg q 6h	Diphenhydramine•HCl 50 mg q 6h
		Ephedrine sulfate 25 mg q 6h
		Monitor blood gases
		Hydrocortisone
		Tracheostomy
REACTION: Bronchospasm		
Epinephrine•HCl 0.3 ml (1:1,000) IM	Epinephrine•HCl 0.3 ml (1:1,000) IM	Oxygen
		Aminophylline 500 mg

Diphenhydramine•HCl 50 mg IV	Aminophylline 250 mg IV over 10-min period of time	IV q 6h Hydrocortisone IV fluids Monitor blood gases Observe for respiratory failure
---------------------------------	--	---

REACTION: Hypotension

Epinephrine•HCl 0.3 ml (1:1,000) IM	Metaraminol bitartrate, 100 mg in 1,000 ml 5% dextrose in water	Oxygen Metaraminol bitartrate IV IV fluids
--	---	---

Transcriber's Notes

- Retained publication information from the printed edition: this eBook is public-domain in the country of publication.
- Corrected a few palpable typos.
- Generated a spine image from elements of the cover image.
- Reformatted several charts for better text flow on narrow screens.
- In the text versions only, text in italics is delimited by underscores.

*** END OF THE PROJECT GUTENBERG EBOOK VENOMOUS
ARTHROPOD HANDBOOK ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the

collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with

this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official

version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, “Information about donations to the Project Gutenberg Literary Archive Foundation.”
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.

- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES

EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a)

distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact

links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility: www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.